

**Cambridge (CIE) IGCSE Maths
Paper 2: Non-Calculator (Set C)
Extended**

Saturday 26 April 2025

Morning (Time: 2 hours 0 minutes)

Total marks

/ 100

Instructions

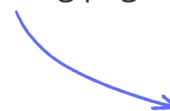
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- Calculators must not be used in this paper.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

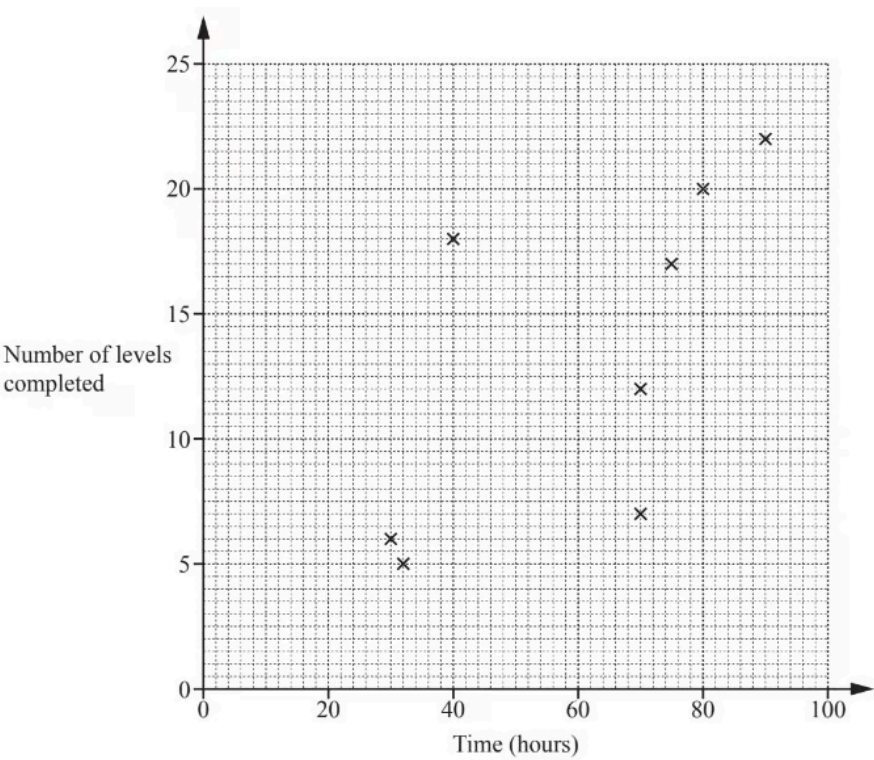
or visit the mock exams landing page for this course



1 (a) Lucy asked 12 people how many hours they each spent playing a computer game and the number of levels they each completed in one month.

The results are shown in the table.

Time spent playing (hours)	90	32	70	75	30	70	40	80	40	65	50	32
Number of levels completed	22	5	12	17	6	7	18	20	8	15	11	9



Complete the scatter diagram.
The first eight points have been plotted for you.

[2]

(2 marks)

(b) One person completes more levels per hour than any of the others.

On the scatter diagram, put a ring around the point for this person.

[1]

(1 mark)

(c) What type of correlation does this scatter diagram show?

[1]

(1 mark)

(d) On the scatter diagram, draw a line of best fit.

[1]

(1 mark)

(e) Another person, Monika, completed 19 levels but forgot to record the time spent playing.

Use your line of best fit to estimate the number of hours that Monika spent playing.

..... hours [1]

(1 mark)

2 Emma records the number of letters in each word in a sentence.

7	1	5	4	2	4	5	3	3	1	2	4
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Find

i) the median,

[2]

ii) the mode,

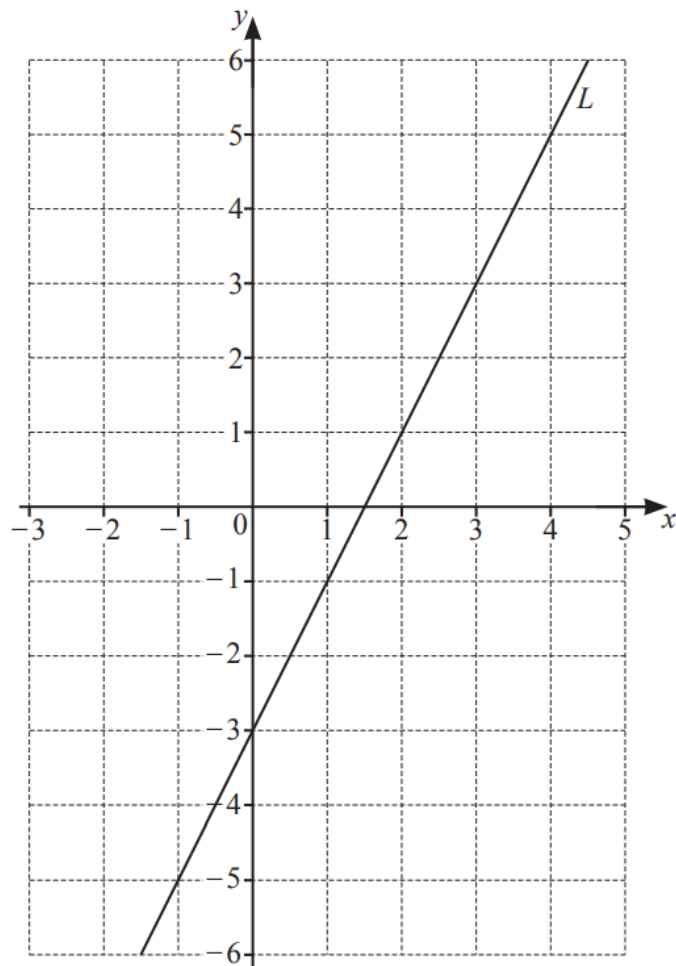
[1]

iii) the range.

[1]

(4 marks)

3



Find the equation of line L in the form $y = mx + c$

$y =$

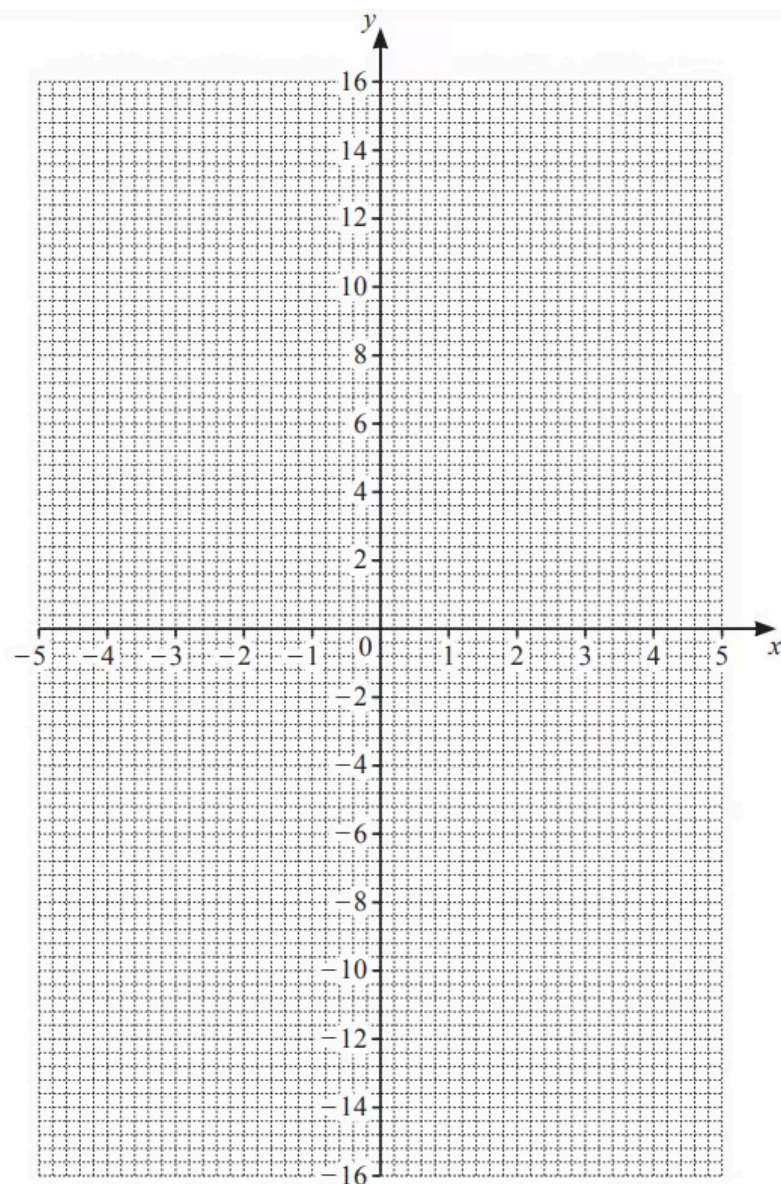
(2 marks)

4 (a) Complete the table of values for $y = \frac{15}{x}$

x	-5	-3	-2	-1		1	2	3	5
y				-15		15			

(3 marks)

- (b) On the grid, draw the graph of $y = \frac{15}{x}$ for $-5 \leq x \leq -1$ and $1 \leq x \leq 5$.



(4 marks)

(c) On the grid, draw the line $y = 6$.

(1 mark)

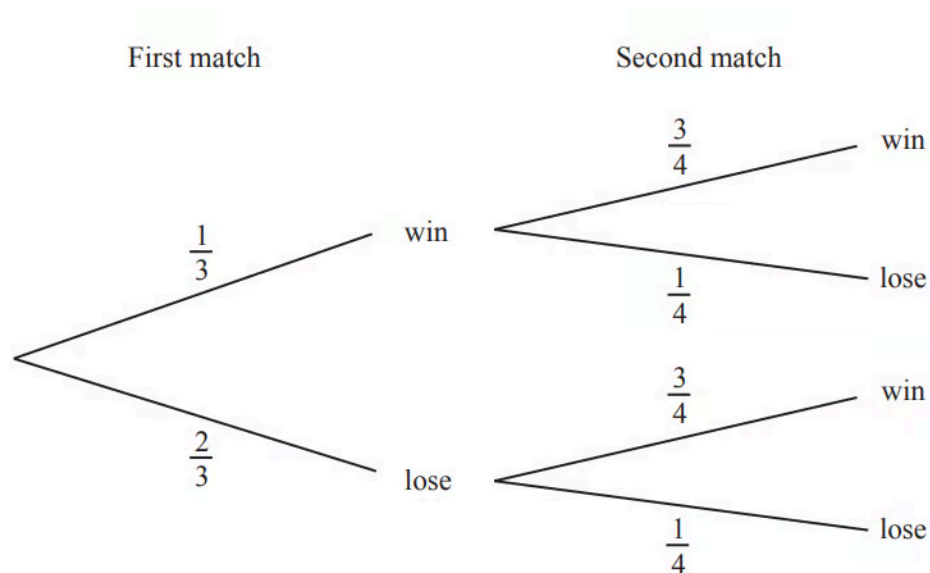
(d) Use your graph to solve $\frac{15}{x} = 6$.

$x = \dots\dots\dots$

(1 mark)

5 A soccer team plays two matches.

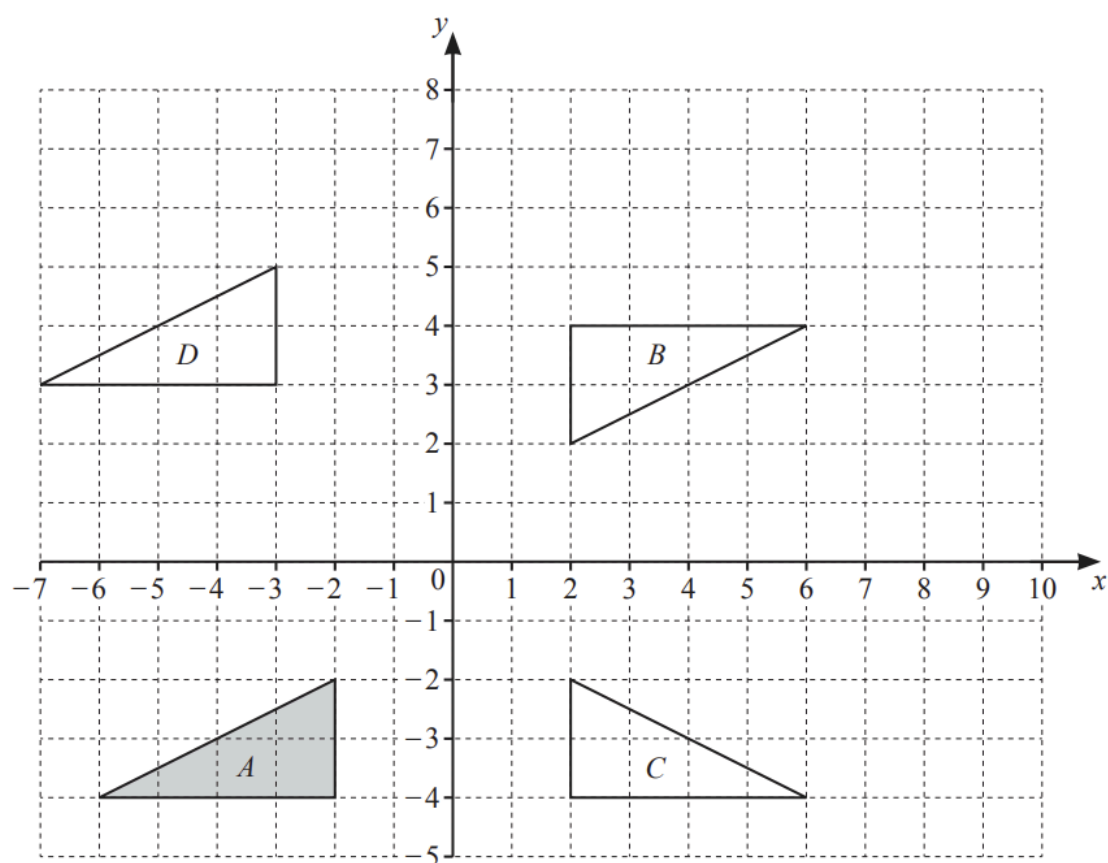
The tree diagram shows the probability of the team winning or losing the matches.



Find the probability that the soccer team wins at least one of the two matches.

(3 marks)

6 (a)



Describe fully the **single** transformation that maps

i) triangle *A* onto triangle *B*,

[3]

ii) triangle *A* onto triangle *C*,

[2]

iii) triangle *A* onto triangle *D*.

[2]

(7 marks)

(b) On the grid, enlarge triangle A by scale factor 0.5, centre (4, 0).

(2 marks)

7 Make p the subject of this formula.

$$H = 7p - 3$$

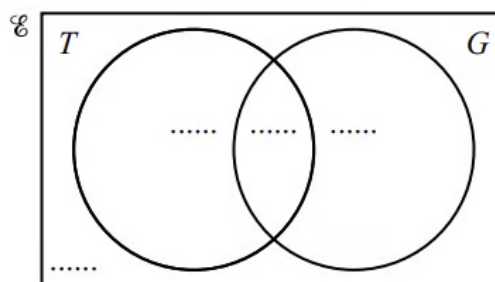
$p = \dots\dots\dots$

(2 marks)

- 8 (a)** $\mathcal{E} = \{\text{children who go to the park}\}$
 $T = \{\text{children who play tennis}\}$
 $G = \{\text{children who play golf}\}$

120 children go to the park.
 50 play tennis.
 75 play golf.
 25 do not play tennis or golf.

Complete the Venn diagram.



[2]

(2 marks)

- (b)** Find $n(T \cap G)$.

(1 mark)

- 9** Solve the simultaneous equations.

You must show all your working.

$$\begin{aligned} 3x - 8y &= 22 \\ x + 4y &= 4 \end{aligned}$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(3 marks)

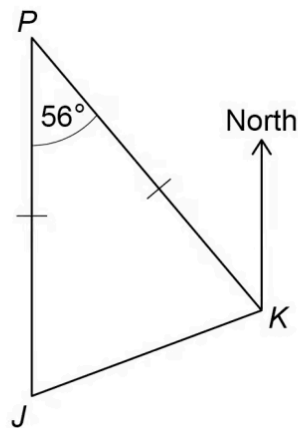
10 J and K are ships.

P is a port.

J is due South of P .

Angle $JPK = 56^\circ$

$JP = KP$



Not drawn
accurately

Work out the bearing of J from K .

(3 marks)

11 $P = 2(w + h)$

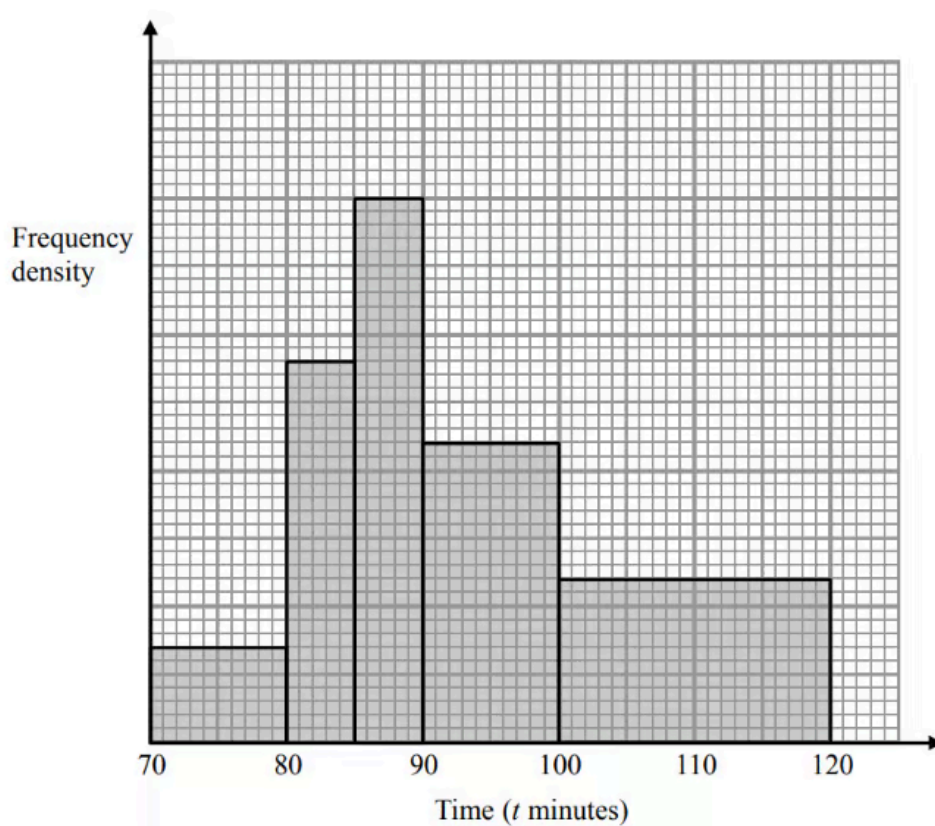
$w = 12$ correct to the nearest whole number.

$h = 4$ correct to the nearest whole number.

Work out the upper bound for the value of P .

(2 marks)

12 The histogram shows information about the time taken by cyclists to finish a cycle race.



7 cyclists took 80 minutes or less to finish the race.

i) Work out an estimate for the number of cyclists who took more than 105 minutes to finish the race.

[3]

ii) Explain why your answer to part (i) is only an estimate.

(4 marks)

13 Rationalise the denominator of $\frac{4}{7 - \sqrt{5}}$

Show each stage of your working.

Give your answer in the form $a + b\sqrt{5}$ where a and b are fractions in their simplest forms.

(3 marks)

14 (a) The table shows the first five terms of sequence A and sequence B .

Term	1	2	3	4	5	6
Sequence A	7	13	23	37	55	
Sequence B	1	3	9	27	81	

Complete the table for the 6th term of each sequence.

(2 marks)

(b) i) Find the n th term of sequence A .

[2]

ii) Find the n th term of sequence B .

[2]

(4 marks)

15 The length of one side of a rectangle is 12cm.

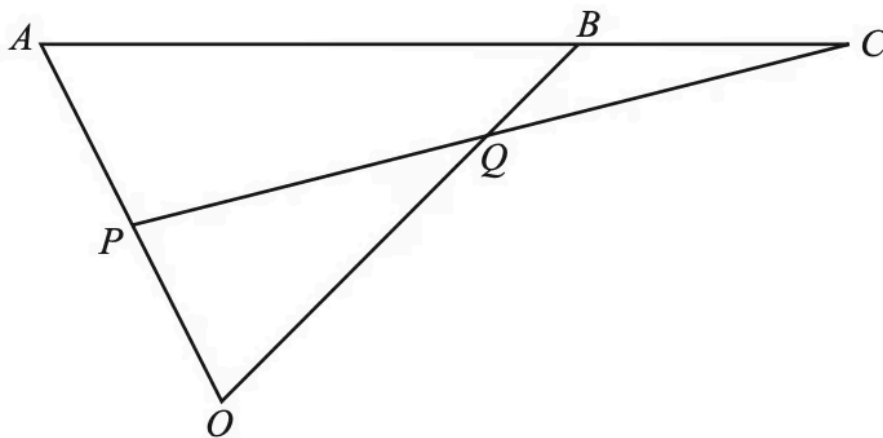
The length of the diagonal of the rectangle is 13cm.

Calculate the area of the rectangle.

..... cm²

(3 marks)

16 (a)



NOT TO
SCALE

OAB is a triangle and ABC and PQC are straight lines.

P is the midpoint of OA , Q is the midpoint of PC and $OQ : QB = 3 : 1$.

$\vec{OA} = 4\mathbf{a}$ and $\vec{OB} = 8\mathbf{b}$.

Find, in terms of $\mathbf{a}$ and/or $\mathbf{b}$, in its simplest form

i) $\vec{AB}$,

$\vec{AB} = \dots\dots\dots [1]$

ii) $\vec{OQ}$,

$\vec{OQ} = \dots\dots\dots [1]$

iii) $\vec{PQ}$,

$\vec{PQ} = \dots\dots\dots [1]$

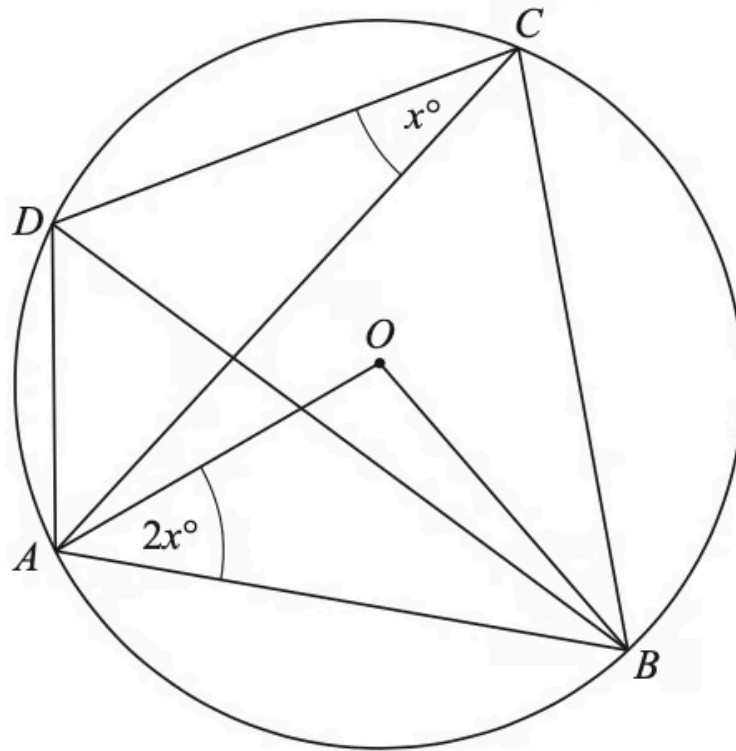
(3 marks)

(b) By using vectors, find the ratio $AB : BC$.

..... :

(3 marks)

17 (a)



NOT TO
SCALE

In the diagram, A, B, C and D lie on the circumference of a circle, centre O .
Angle $ACD = x^\circ$ and angle $OAB = 2x^\circ$.

Find an expression, in terms of x , in its simplest form for angle AOB .

Angle $AOB = \dots\dots\dots$

(1 mark)

(b) Find an expression, in terms of x , in its simplest form for angle ACB .

Angle $ACB = \dots\dots\dots$

(1 mark)

(c) Find an expression, in terms of x , in its simplest form for angle DAB .

Angle $DAB = \dots\dots\dots$

(2 marks)

18 Work out $\frac{5}{6} \div 1\frac{1}{3}$.

You must show all your working and give your answer as a fraction in its simplest form.

(3 marks)

19 $ABCDEFGHI$ is a regular 9-sided polygon.

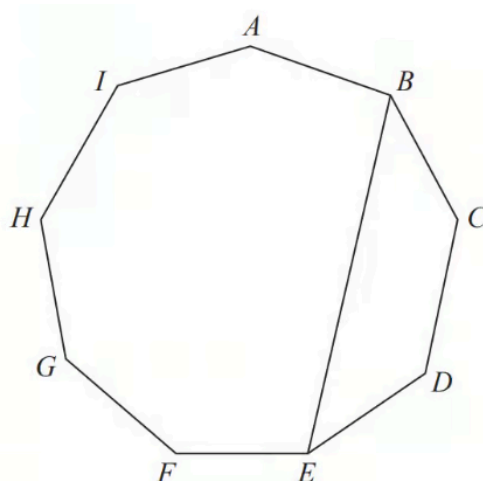


Diagram **NOT**
accurately drawn

The vertices B and E are joined with a straight line.

Work out the size of angle BEF .

You must show how you get your answer.

(4 marks)

20 Find the value of $\left(\frac{1}{81}\right)^{-\frac{3}{4}}$.

(2 marks)

21 Simplify.

$$\frac{ux - 2u - x + 2}{u^2 - 1}$$

(4 marks)

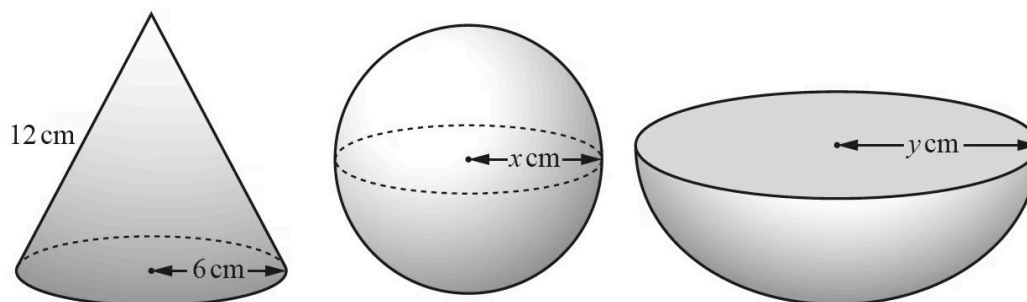
22

$$g(x) = 1 - 2x$$

Find the value of

Find $g(x)g(x) - gg(x)$, giving your answer in the form $ax^2 + bx + c$.

(4 marks)



The diagram shows three solids.

The base radius of the cone is 6 cm and the slant height is 12 cm.

Show that the total surface area of the cone is $108\pi \text{ cm}^2$.

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi rl$.]

(2 marks)

- 24** The radius of the sphere is x cm and the radius of the hemisphere is y cm.
The **total** surface area of each solid is the same.

Find the value of x and the value of y .

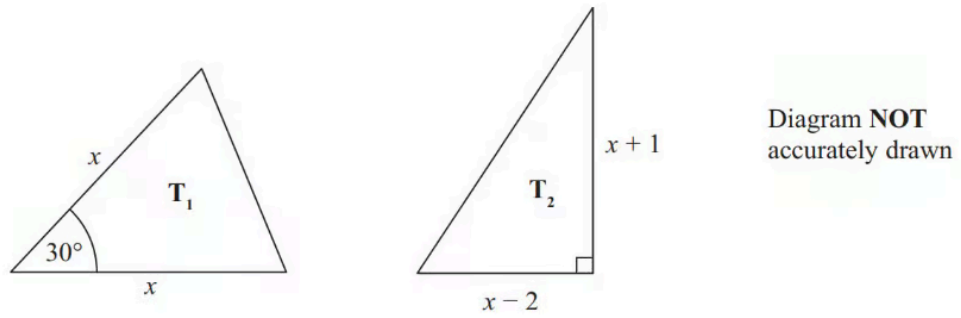
Give the value of x as a simplified surd.

[The surface area, A , of a sphere with radius r is $A = 4\pi r^2$.]

$x = \dots\dots\dots$

$y = \dots\dots\dots$

(4 marks)



The lengths of the sides are in centimetres.

The area of triangle T_1 is equal to the area of triangle T_2 .

Work out the value of x , giving your answer in the form $a + \sqrt{b}$ where a and b are integers.

(5 marks)